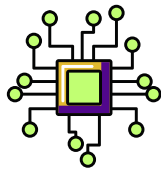


Robustness of Quantum Teleportation under Bell Entangled Channel.

Noise & Routing-overhead Study with Error Mitigation via Linear ZNE



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Abstract

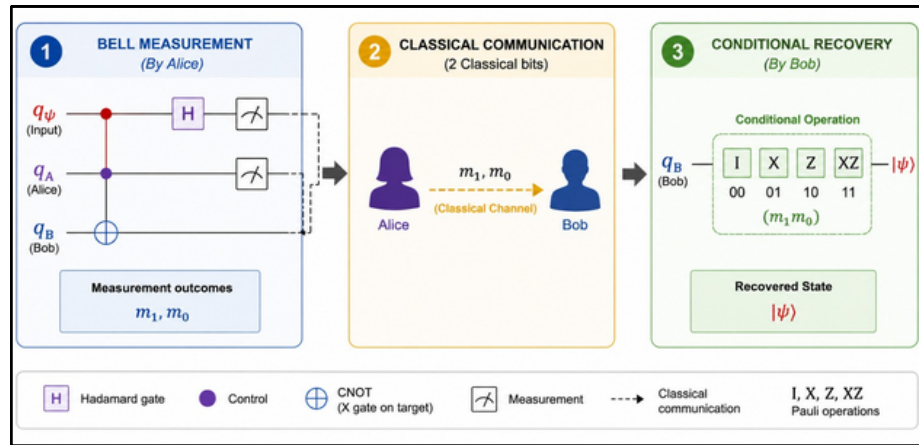
Teleportation is a core primitive for quantum networks, but fidelity degrades on NISQ hardware (via noise channels and compiler-inserted SWAP routing) with unclear relative impact. We simulate 4 noise channels and natural vs. forced-SWAP layouts on real IBM calibration data (FakeSherbrooke), then apply Linear ZNE to mitigate the worst case. Bit/phase-flip prove most damaging; forced routing causes a significant 1.40% fidelity drop (vs. 2.33% predicted), revealing real circuit errors partially cancel rather than compound worst-case. ZNE recovers ~2.2 percentage points, showing software-level mitigation can offset hardware-level routing cost - a practical path toward NISQ-viable quantum communication.

Introduction

Protocol & Bell states

- Transfers unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, Alice $\rightarrow$ Bob, via shared Bell state + 2 classical bits

$$|\Phi_{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, |\Psi_{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$



Noiseless case (ideal):

$$F = \langle \psi | \rho_{out} | \psi \rangle = 1$$

Performance Metric

$$F = \langle \psi | \rho_{out} | \psi \rangle, \quad F_{classical} = \frac{2}{3}$$

Noise formalism (Kraus operators)

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \quad \sum_k E_k^\dagger E_k = I.$$

Noisy-Teleportation Reduction [1]

$$E(\rho) = (1-p)\rho + pA\rho A, \quad A \in X, Y, Z$$

$$\mathcal{E}(\rho_{out}) = (1-p)\rho_{in} + pA\rho_{in}A \Rightarrow F(p) = (1-p) + p\langle A \rangle^2$$

Routing-Induced Noise.

SWAP-as-3-CNOTs identity:

$$|a, b\rangle \rightarrow |a, a \oplus b\rangle \rightarrow |a \oplus (a \oplus b), a \oplus b\rangle = |b, a \oplus b\rangle \rightarrow |b, (a \oplus b) \oplus b\rangle = |b, a\rangle$$

- Confirms one logical SWAP = 3 real hardware gates

$$SWAP(a, b) = CNOT(a, b) \cdot CNOT(b, a) \cdot CNOT(a, b)$$

- Approximate Effect of Induced Noisy gates: $F \approx \prod_i (1 - \varepsilon_i)$

Error Mitigation.

- Zero Noise Result: Extrapolating to $\lambda \rightarrow 0$ via the Richardson two-point formula:

$$F_{ZNE} = \frac{3F(1) - F(3)}{2}.$$

Methodology

Stage 1 (Ideal Baseline).

- Full 3-qubit teleportation circuits built & simulated via exact density-matrix evolution (no shot noise)
- Hand-derived check against fixed test state

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

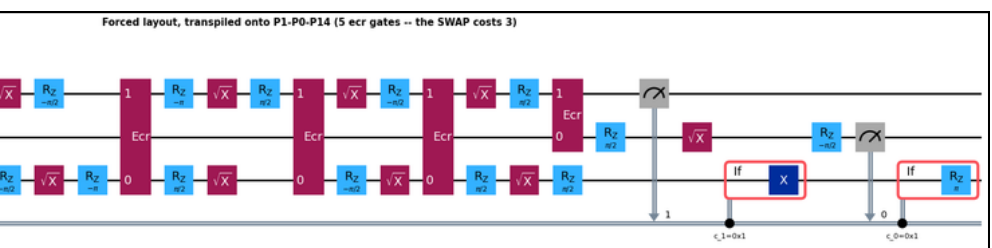
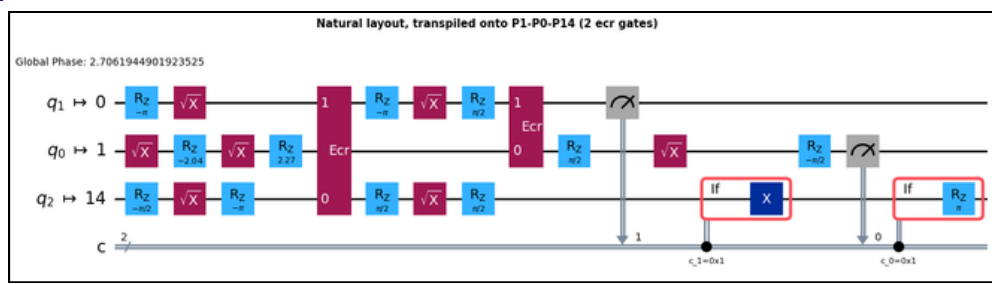
Stage 3 - Routing-Induced Noise (Natural vs. SWAP).

- Forced layout:** Alice/Bob on end qubits $\rightarrow$ 1 SWAP (3 CNOTs $\rightarrow$ forced = 5 two-qubit gates total)

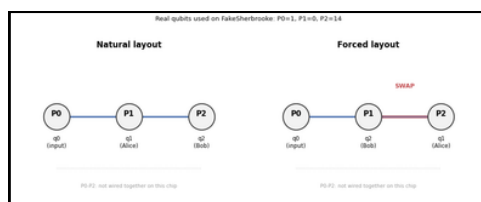
Stage 2 (Noise-Channel Sweep) $p \in [0, 0.75]$

- Channels:** bit-flip, phase-flip, depolarizing (diagonal Pauli), amplitude-damping (non-Pauli, non-unital/diagonal)
- $n = 150$ Haar-random input states per (channel, p) [5]
- Bit/phase/depolarizing: noise on input qubit = on Bell-pair noise [2]
- Amplitude-damping: Not equal hence noise on bell pair. [2]
- Qiskit, Hand Driven, QuTip - Triple cross-check: all agree to $< 10^{-6}$

- 2Q gates only between adjacent physical qubits
- Natural layout:** 2 CNOTs, (no SWAP)



- Transpiled onto FakeSherbrooke (real 127-qubit calibration snapshot)
- Fidelity: calibration-only $F \approx \prod(1 - \varepsilon_i)$ [6], vs. full noisy sim (40 states $\times$ 4 Bell $\times$ 2 layouts $\times$ 2048 shots)



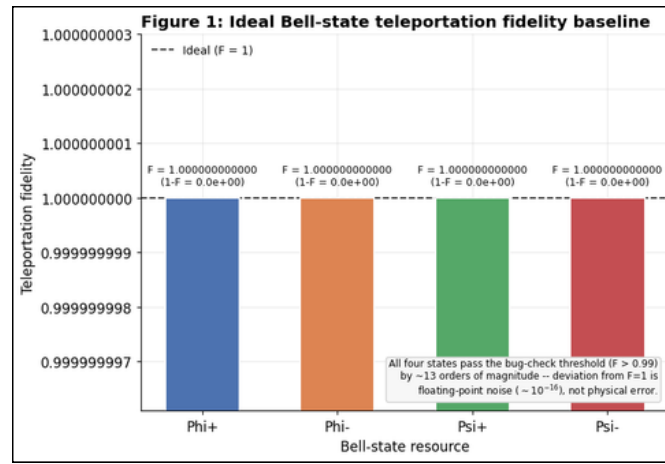
Stage 4 - Linear ZNE Mitigation.

- Target: worst Stage 3 case (Ψ^- , forced layout)
- Gate folding $U \rightarrow U(U^\dagger U)^k$, ideal no-op, scales real noise [4]
- Linear Richardson extrapolation with quadratic Lagrange check to calculate Recovery.

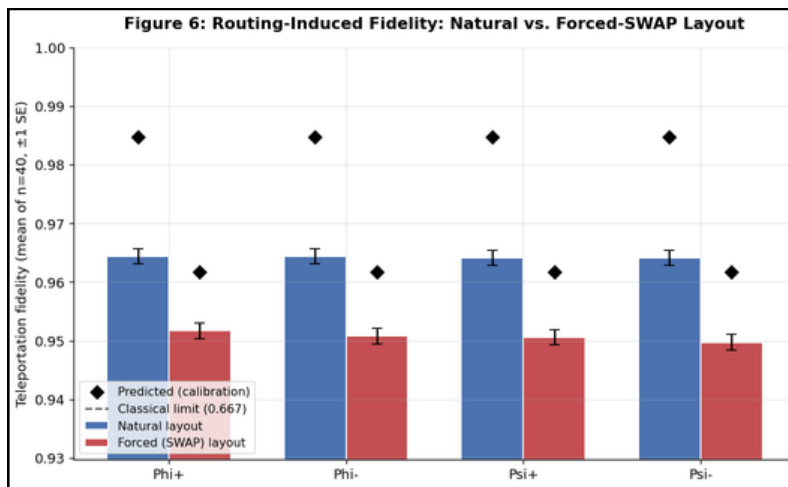
Results & Discussions

STAGE 1:

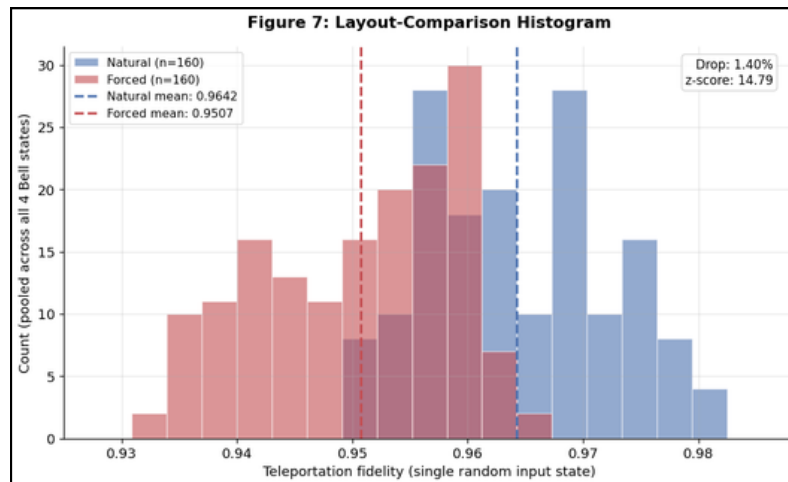
All states achieve $F=1$ ($\sim 10^{-16}$), confirming the circuit logic is correct prior noise



Stage 3 - Routing Overhead.

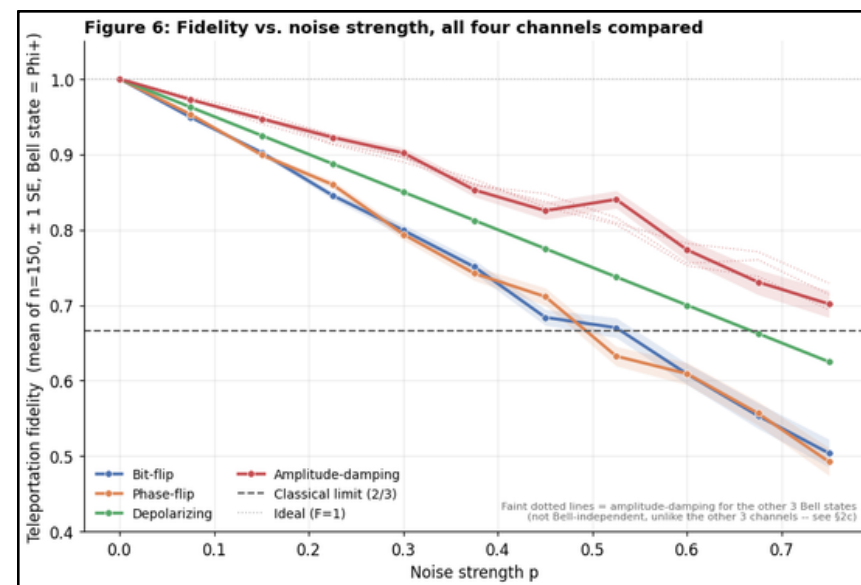


Predicted drop: 2.33% (calibration-only via FakeSherbrooke, $F = \prod(1 - \varepsilon_i)$). Actual (simulation): 1.40% ($z=14.8$, far past the $z>2$ significance threshold). The calibration model overstates the loss because it is a pessimistic worst-case bound: it assumes every gate error compounds independently. In a structured circuit, real errors are partly coherent and can partially cancel, so the observed drop falls below that worst-case prediction. All four Bell states perform nearly identically within each layout. The lowest-fidelity case is Ψ^- on the forced layout, selected for Stage 4



STAGE 2: Abstract Noise channel ranking (at $p = 0.75$)

Noise applied to input qubit: Bit-flip, phase-flip, and depolarizing yield identical fidelity across all states. Amplitude-damping doesn't: applied to the Bell pair, it shows a modest Bell-state-dependent spread (~ 0.03 at $p=0.75$).



Phase-flip and bit-flip are most damaging because they push Haar-random states along a single fixed axis; depolarizing spreads error across three axes, allowing partial cancellation, while amplitude-damping preserves the $|0\rangle$ fixed point.

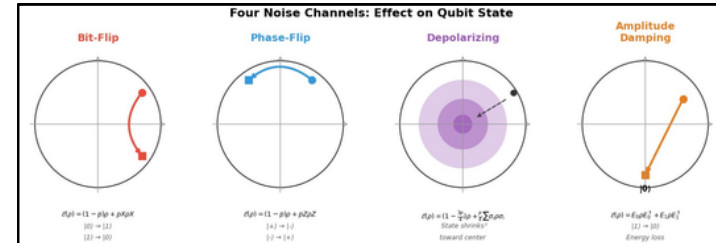
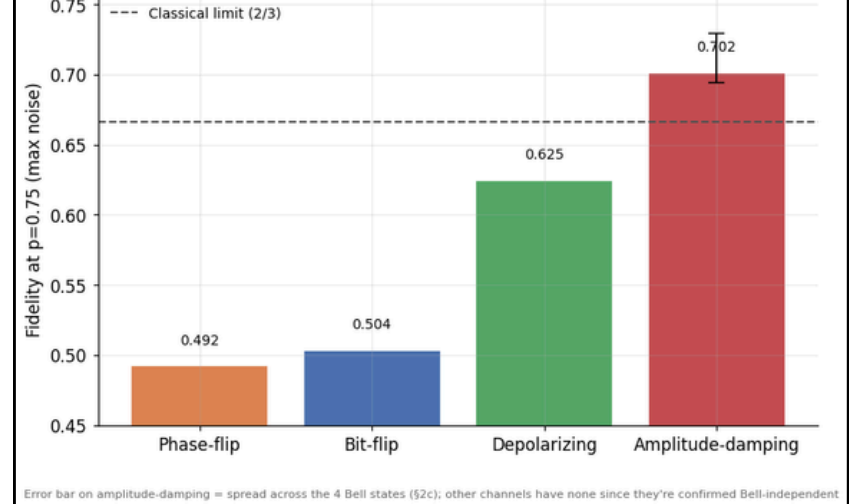


Figure 7: Channel ranking at maximum noise strength

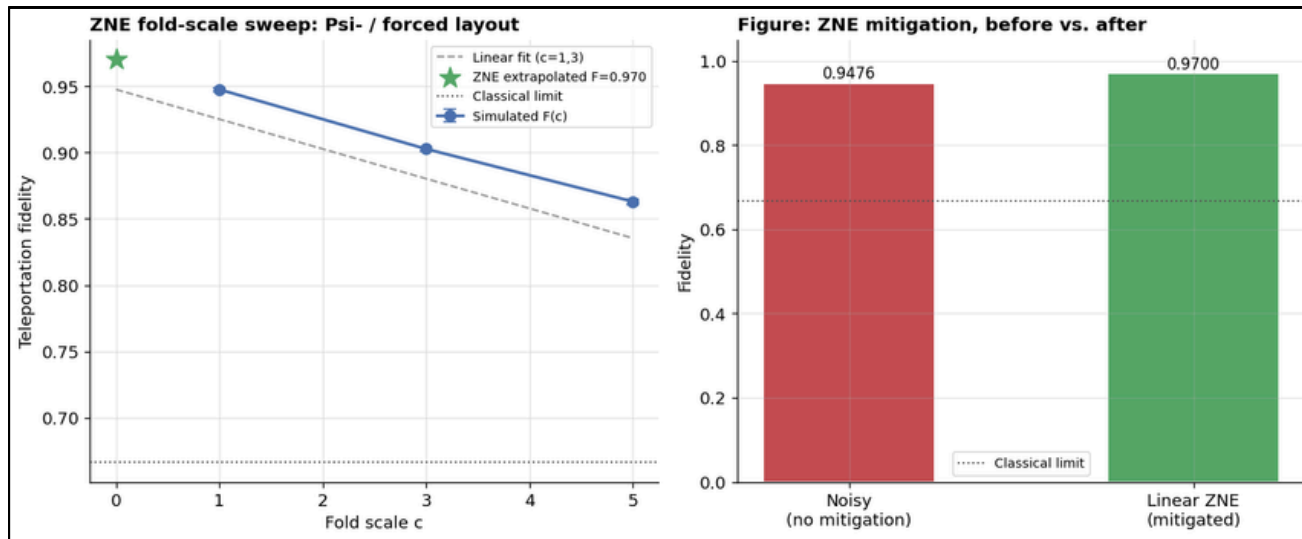


Stage 4 - ZNE Recovery.

Linear extrapolation gives $F_{ZNE}=0.9700$; quadratic Lagrange gives 0.9719 ($< 0.2\%$ difference), confirming no overshoot and no polynomial fallback needed. This recovers ~ 2.2 percentage points of fidelity lost to the forced layout's SWAP overhead.

λ	Fidelity
1 $\times$ (raw)	0.9476
3 $\times$	0.9027
5 $\times$	0.8631

The mitigated value (0.9700) sits slightly above the natural-layout average (0.9642) because ZNE removes the forced layout's extra two-qubit gate error, while the natural layout still carries its own unmitigated two-qubit error plus the shared baseline noise.



Future Plans

- NISQ devices remain routing-constrained: as circuits scale, compiler-inserted SWAPs are one of the largest controllable noise sources - this is an active bottleneck in near-term quantum hardware.
- Teleportation-based routing is being explored as an alternative to SWAP-heavy compilation in circuit mapping [3]
- SWAP overhead and its mitigation are directly relevant to quantum repeater and quantum-internet hardware design
- Validating these results on real hardware (and extending them to multi-hop, network-scale teleportation) is a direct step toward the noise-resilient quantum communication links required for a practical quantum internet.

Conclusion

Routing overhead (not bell state) dominates hardware-limited teleportation errors, though it remains smaller in practice than calibration predictions. Linear ZNE successfully recovers ~ 2.2 percentage points of fidelity lost to SWAP-induced noise, validating its practical utility as a complement to hardware connectivity constraints on near-term (NISQ) devices.

References

- [1] Oh, S., Lee, S., & Lee, H. (2002). Fidelity of quantum teleportation through noisy channels. Phys. Rev. A, 66, 022316.
- [2] Usher, N., & Browne, D. E. (2017). Noise in one-dimensional measurement-based quantum computing. Quantum Inf. Comput., 17, 1372-1397. (arXiv:1704.07298)
- [3] Hillmich, S., Zulehner, A., & Wille, R. (2021). Exploiting quantum teleportation in quantum circuit mapping. Proc. ASP-DAC '21, 792-797. (arXiv:2011.07314)
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- [6] T. Rindell, B. Yenilen, N. Halonen, A. Pönni, I. Tittonen, and M. Raasakka, "Exploring the optimality of approximate state preparation quantum circuits," Physics Letters A, vol. 475, p. 128860, 2023 (arXiv:2210.06411)